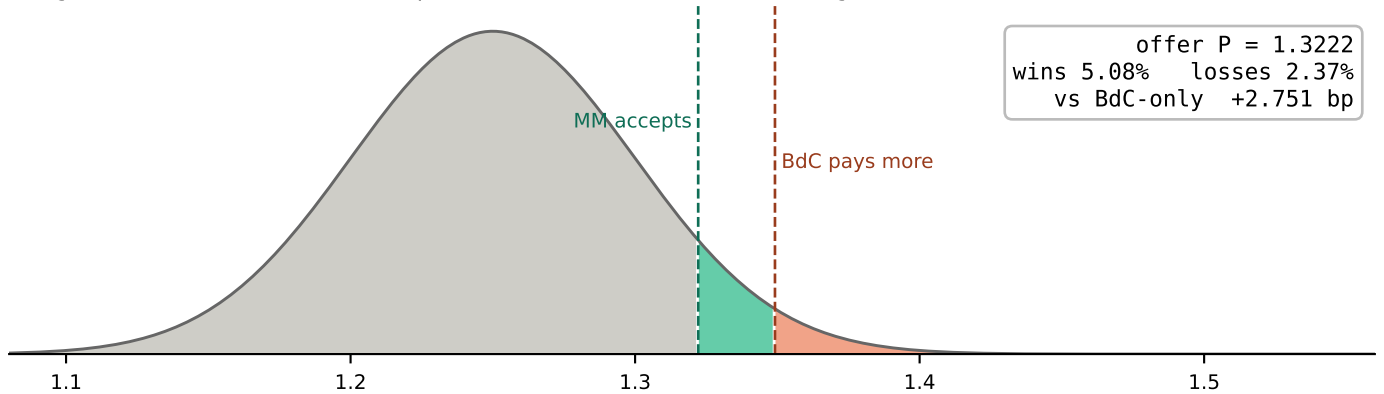


Why the MM's fee collapses the one-shot market

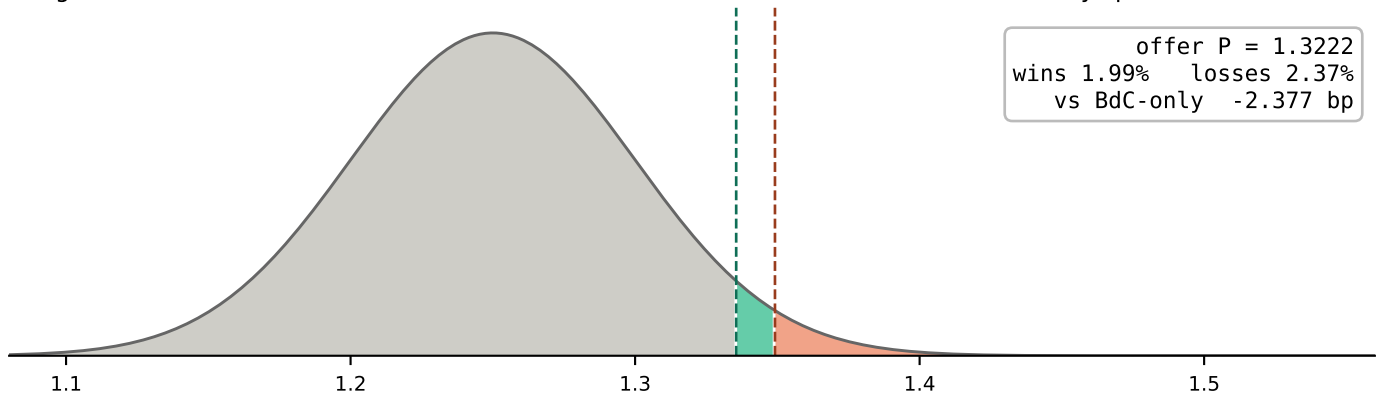
Side A (GBP→USD) · true rate $X \sim N(1.25, 0.05)$ · trader BdC fee $f = 2\%$ · MM BdC fee g

The trader writes an offer P before X is drawn. The MM accepts iff $P < (1-g)X$; otherwise the trader takes the BdC at $0.98X$. So a fixed P wins only when X lands in the strip $P/(1-g) < X < P/0.98$. Below the strip: rejected, no harm. Above it: accepted, but the BdC would have paid more — a loss that grows with X . The whole question is where to put the strip.

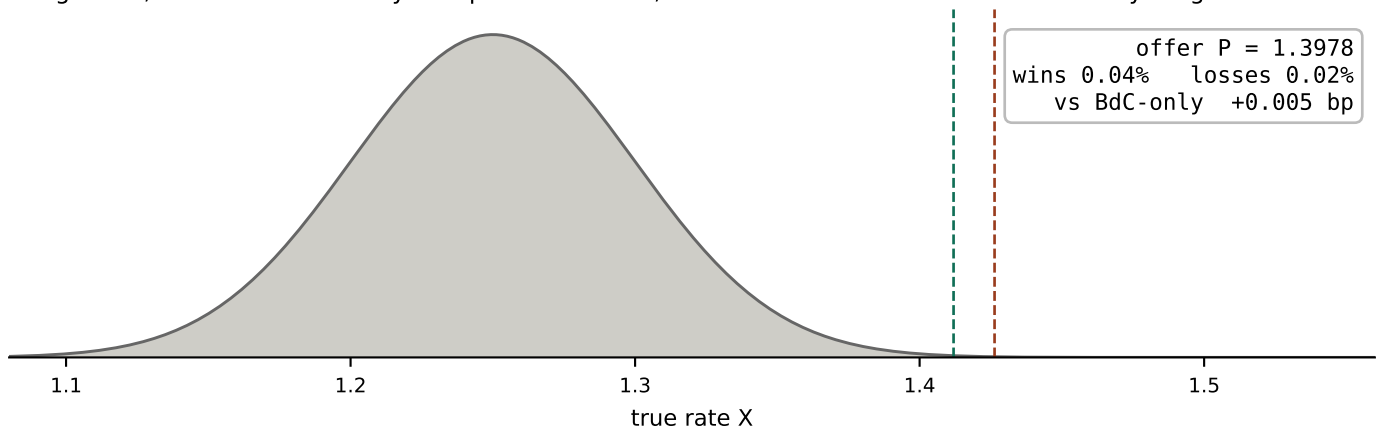
1. $g = 0$, best offer. The 2% strip sits 1.4 sd out: wins 5.1% outweigh losses 2.4%.



2. $g = 1\%$, same offer. The MM's line rises; the BdC line is the trader's own fee and stays put.



3. $g = 1\%$, best offer. The only escape is 3.2 sd out, where losses are rare — but so is everything.



Halving the strip does not halve the profit. It roughly doubles how far out the strip must sit, and the chance of X landing that far out falls by a factor of about a hundred ($7.4\% \rightarrow 0.06\%$). The gain over the BdC goes $+2.75 \text{ bp} \rightarrow +0.005 \text{ bp}$: the optimum still exists but is worth nothing, so the payoff is flat in P and nothing pins the co-trained offer down — it drifts instead of converging. A Monte Carlo re-optimisation of this exact model (400,000 common-random-number draws) reproduces the closed form at $g = 0$ and stops resolving an optimum at the same fee.